

Nate Cirino

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Available May 2027

EDUCATION

Northeastern University

May 2027 | MGPA: 3.6/4.0

B.S. in Computer Science, Systems Concentration

Boston, MA

Coursework: Computer Systems, Digital Design & Computer Architecture, Computer Networks, Algorithms & Data Structures, Object-Oriented Design, Foundations of Software Engineering, Systems Security, Cybersecurity, Foundations of Data Science

PROJECTS

NOAA AUV Sonar Detection | *Python, PyTorch, AWS Neuron SDK, SageMaker*

April 2025 – July 2025

- Built onboard detection software for NOAA autonomous underwater vehicles, classifying novel aquatic signatures from thermal sensor spectrograms with CNN models trained on real mission data
- Designed AWS Neuron SDK deployment path for onboard custom-silicon inference under hard real-time limits, targeting resource-constrained compute on the vehicle
- Refined architecture and loss formulation to address class imbalance and raw sensor noise, improving detection

Kepler | *Rust, C++, Raft, LSM-tree, MVCC, Tokio*

Mar. 2026 – Present

- Built a distributed, linearizable key-value store in Rust with from-scratch Raft consensus and LSM-tree engine
- Implemented Raft election, log replication, and membership changes; storage: memtable, WAL, leveled SSTables
- Wrote concurrent async node runtime on Tokio with an MVCC transaction layer giving snapshot isolation and linearizable reads across a multi-node cluster, validated by deterministic correctness tests
- Built single-tier compaction with full SSTable rewrite every 4 flushes; inline, writer-blocking execution

Sentinel | *Next.js, FastAPI, Claude, pgvector, OpenTelemetry, Docker*

Apr. 2026 – Present

- Built an AI SRE agent autonomously triaging incidents via tool-use over Prometheus, Tempo, and Loki telemetry
- Built pgvector RAG over runbooks and postmortems with prompt caching on frozen schemas; 100% eval accuracy
- Human-in-the-loop remediation gated behind approval; running live in SculptAI production triaging incidents

EXPERIENCE

Wayfair

Jul 2026 – Present

Software Engineering Co-op

Boston, MA

- Own front-of-house device stack across all Wayfair physical retail brands (AllModern, Birch Lane, Joss & Main, Perigold): e-label hardware, mobile, and customer/associate-facing POS
- Building a supplier-agnostic e-label API unifying 4 hardware suppliers behind one interface, powering new outlet launches; debug device- and integration-level failures across the software/hardware boundary
- Shipped POS checkout flows: in-store scan-to-basket customer switching and live associate-entry customer sync

Wolters Kluwer

Jul 2025 – Jun 2026

Software Engineer

Boston, MA

- Overhauled Relay JMS broker serialization with json-io across 20+ microservices, eliminating timeouts at scale
- Optimized Expert AI search and filtering across 20+ microservices, improving clinical query relevance at scale
- Built Java Spring microservice for CME credit issuance inside Expert AI's clinician-facing workflow
- Integrated Expert AI into CME accrual/redemption microservices, expanding GenAI CDS to 250K+ clinicians
- Implemented OIDC/IdP authentication via Azure across 20 enterprise health systems, securing 10K+ clinicians

SculptAI

Jan 2025 – Aug 2026

Software Architect

New York, NY

- Maintained 99.99% uptime across 1-2M monthly model calls for 2K paying users via AWS WAF, Datadog, and K8s
- Architected Flask backend with RAG pipeline for personalized workout recommendations using Bedrock KB
- Led eng org restructure at \$1M-funded startup, partitioning teams into DevOps, dev, and agentic workflows; drove architecture tradeoffs across a cross-functional team and continue as technical advisor

Chevron New Energies

May 2024 – Aug 2024

Software Engineer Intern

Houston, TX

- Implemented CI/CD workflows using Azure DevOps and K8s, enabling automated deployment and rollback
- Built RESTful APIs and data pipelines using Azure Data Factory, improving product efficiency by 40%

SKILLS & HONORS

Languages/Tools: Rust, C++, C, Python, Java, TypeScript, JavaScript, Bash, Linux, Docker, Kubernetes, Terraform, Jenkins, Buildkite, Groovy, AWS, Azure, PostgreSQL, GraphQL

Frameworks & Libraries: Tokio, PyTorch, TensorFlow, Pandas, FastAPI, Flask, Node.js, Express, React, Next.js

AI / Agentic Tooling: Devin, Cursor, Claude Code, Claude Cowork, GitHub Copilot, CodeRabbit

Honors: Tech Lead at Northeastern Electronic Racing (student engineering design team), Dean's List (All Semesters), json-io OSS Contributor, PawHacks